

EC614: Macroeconomic Forecasting

School of Economics, QAU, Islamabad

Section A: Multiple Choice Questions (25 marks)

Select the best answer for each question.

Question 1

In a reduced-form VAR, the error terms across equations are:

- a) Always uncorrelated by construction
- b) May be correlated if variables are related
- c) Always normally distributed
- d) Independent of past values

Question 2

The primary advantage of ARDL over traditional cointegration methods is:

- a) It requires all variables to be $I(1)$
- b) It works with variables of mixed integration orders $I(0)$ and $I(1)$
- c) It is computationally simpler
- d) It doesn't require lag selection

Question 3

In Granger causality testing, if X Granger-causes Y, it means:

- a) X definitely causes Y in a causal sense
- b) Past values of X help predict Y beyond Y's own past
- c) Y cannot Granger-cause X simultaneously
- d) X and Y must be cointegrated

Question 4

The Pesaran-Shin-Smith (2001) bounds test in ARDL tests the null hypothesis:

- a) All variables are $I(1)$
- b) Variables are cointegrated
- c) No long-run relationship exists
- d) Variables are stationary

Question 5

In a structural VAR (SVAR), identification restrictions are imposed on:

- a) The lag length only
- b) The contemporaneous relationships among variables
- c) The forecast horizon
- d) The variance of residuals

Question 6

Which statement about impulse response functions (IRFs) is FALSE?

- a) They trace dynamic effects of shocks over time
- b) They require the VAR to have uncorrelated errors
- c) Confidence bands indicate statistical uncertainty
- d) They are always symmetric between variables

Question 7

In a recursive VAR, the ordering of variables matters because:

- a) It determines computational efficiency
- b) It imposes assumptions about contemporaneous effects
- c) It affects lag length selection
- d) It changes the sample size

Question 8

The forecast error variance decomposition (FEVD) answers:

- a) How accurate the forecast is
- b) What fraction of forecast error variance is due to each shock
- c) Whether the model is correctly specified
- d) If variables are cointegrated

Question 9

If two $I(1)$ variables are cointegrated, their linear combination is:

- a) $I(1)$
- b) $I(2)$
- c) $I(0)$
- d) Non-stationary with no defined order

Question 10

The error correction term in a VECM represents:

- a) The forecasting error
- b) Deviation from long-run equilibrium
- c) Short-run dynamics
- d) Measurement error

Question 11

In the Johansen cointegration test, the trace statistic tests:

- a) Whether $\text{rank}(\Pi) = 0$
- b) Whether $\text{rank}(\Pi) \leq r$ against $\text{rank}(\Pi) > r$
- c) Stationarity of individual series
- d) Granger causality

Question 12

Weak exogeneity in a VECM context means:

- a) The variable doesn't respond to disequilibrium
- b) The variable is determined outside the model
- c) The variable is stationary
- d) The variable has weak predictive power

Question 13

The primary problem with estimating a VAR with non-stationary I(1) variables that are NOT cointegrated is:

- a) Computational complexity
- b) Spurious regression
- c) Loss of degrees of freedom
- d) Heteroskedasticity

Question 14

In ARDL modeling, if the F-statistic falls between the lower and upper bounds:

- a) We reject cointegration
- b) We accept cointegration
- c) The result is inconclusive
- d) We need to increase lag length

Question 15

The Cholesky decomposition in recursive VAR estimation:

- a) Tests for cointegration
- b) Orthogonalizes the VAR residuals
- c) Selects optimal lag length
- d) Estimates structural parameters

Question 16

Compared to univariate ARIMA models, VARs:

- a) Are always superior for forecasting
- b) Capture cross-variable dynamics
- c) Require less data
- d) Have clearer structural interpretation

Question 17

In panel data with fixed effects, the "within" transformation:

- a) Adds time-specific effects
- b) Removes individual-specific means
- c) Differences all variables
- d) Standardizes variables

Question 18

The Hausman test in panel data compares:

- a) Fixed effects vs pooled OLS
- b) Random effects vs pooled OLS
- c) Fixed effects vs random effects
- d) Static vs dynamic models

Question 19

In SVAR identification using the Blanchard-Perotti approach:

- a) Only short-run restrictions are used
- b) Institutional knowledge provides parameter values
- c) All variables must be ordered recursively
- d) Long-run restrictions are necessary

Question 20

When VAR residuals show autocorrelation, the most likely issue is:

- a) Too many lags included
- b) Insufficient lag length
- c) Non-stationarity
- d) Heteroskedasticity

Part B: Short Answer Questions (30 marks)

Answer all questions. Write concisely but completely.

Question 1 (5 marks)

Explain the difference between reduced-form VAR, recursive VAR, and structural VAR. Under what circumstances would you prefer using each type? Provide a concrete example from macroeconomic policy analysis.

Question 2 (5 marks)

You estimate an ARDL(2,4,0,0,0,4,4,1) model for a relationship between multiple variables.

- (a) What do the numbers in parentheses represent?
- (b) The bounds test F-statistic = 6.45 (critical values: $I(0) = 2.45$, $I(1) = 3.61$ at 5% level). What do you conclude about cointegration?
- (c) If cointegration exists, what is your next step in the analysis?

Question 3 (5 marks)

Consider a three-variable VECM for money (M), prices (P), and GDP (Y) for Pakistan.

The Johansen test indicates $r = 1$ cointegrating relationship. The estimated cointegrating vector is:

$$\beta' = [1, -0.85, -0.35] \text{ for } [M, P, Y]$$

- (a) Interpret this long-run relationship economically.
- (b) What does $r = 1$ tell you about the system?
- (c) How would you test if money (M) is weakly exogenous?

Question 4 (15 marks)

You are analyzing monetary policy transmission in Pakistan using quarterly data from 2000Q1 to 2023Q4 for the following variables:

- GDP growth rate (Y)
- CPI inflation (π)
- Policy interest rate (R)
- Real effective exchange rate (REER)

Part (a): Preliminary Analysis (5 marks)

- (i) Explain the steps you would take to prepare the data before estimating a VAR model.
- (ii) What unit root tests would you apply and why?
- (iii) How would you handle the different orders of integration if some variables are $I(0)$ and others are $I(1)$?

Part (b): VAR Specification (5 marks)

- (i) Explain how you would select the optimal lag length for the VAR.
- (ii) Would you include deterministic terms (constant, trend)? Justify your answer.
- (iii) Write out the VAR equations in matrix form for lag order $p=2$.

Part (c): Structural Identification (5 marks)

- (i) Propose a recursive ordering for these variables based on institutional knowledge of Pakistan's economy. Justify your ordering.
- (ii) Alternatively, propose one short-run restriction and one long-run restriction for SVAR identification.
- (iii) What economic shock are you trying to identify, and why does it matter for policy?